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FURTHER MATHEMATICS

9231/32

Paper 3 Further Mechanics

October/November 2023

1 hour 30 minutes

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 ms^{-2} .

INFORMATION

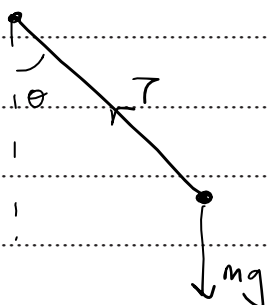
- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

This document has **16** pages. Any blank pages are indicated.

- 1 One end of a light inextensible string of length a is attached to a fixed point O . The other end of the string is attached to a particle of mass m . The string is taut and makes an angle θ with the downward vertical through O , where $\cos \theta = \frac{2}{3}$. The particle moves in a horizontal circle with speed v .

Find v in terms of a and g .

[4]



$$T \cos \theta = mg$$

$$T \sin \theta = \frac{mv^2}{r}$$

$$\tan \theta = \frac{v^2}{rg}$$

$$r = a \sin \theta$$

$$v^2 = \left(\frac{\sin \theta}{\cos \theta} \right) (a \sin \theta) (g)$$

$$= \left(\frac{1 - \cos^2 \theta}{\cos \theta} \right) ag$$

$$v = \sqrt{\frac{5}{6} ag}$$

- 2 A particle P of mass 0.5 kg moves in a straight line. At time $t \text{ s}$ the velocity of P is $v \text{ ms}^{-1}$ and its displacement from a fixed point O on the line is $x \text{ m}$. The only forces acting on P are a force of magnitude $\frac{150}{(x+1)^2} \text{ N}$ in the direction of increasing displacement and a resistive force of magnitude $\frac{450}{(x+1)^3} \text{ N}$. When $t = 0$, $x = 0$ and $v = 20$.

Find v in terms of x , giving your answer in the form $v = \frac{Ax+B}{(x+1)}$, where A and B are constants to be determined. [6]

$$a = \frac{300}{(x+1)^2} - \frac{900}{(x+1)^3}$$

$$v \frac{dv}{dx} = \frac{300}{(x+1)^2} - \frac{900}{(x+1)^3}$$

$$\int v dv = \int \frac{300}{(x+1)^2} - \frac{900}{(x+1)^3} dx$$

$$\frac{v^2}{2} + C = \frac{-300}{x+1} + \frac{450}{(x+1)^2} \quad \Rightarrow \text{when } x=0, v=20$$

$$200 + C = -300 + 450$$

$$\rightarrow C = -50$$

$$\frac{v^2}{2} = \frac{450}{(x+1)^2} - \frac{300}{x+1} + 50$$

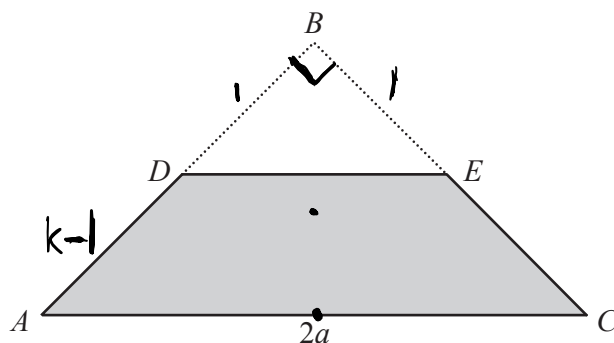
$$v^2 = \frac{900 - 600(x+1) + 100(x+1)^2}{(x+1)^2}$$

$$v^2 = \frac{400 - 400x + 100x^2}{(x+1)^2} = \frac{(10x-20)^2}{(x+1)^2}$$

$$v = \frac{\pm(10x-20)}{x+1}$$

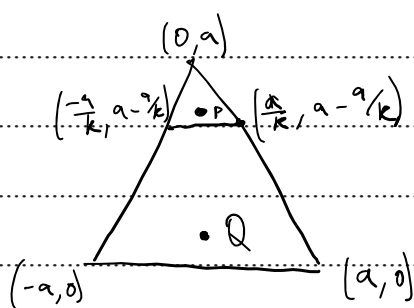
substituting $\rightarrow 20 = \frac{10(0)-20}{1} \Rightarrow$ we take -ve root

$$v = \frac{20-10x}{x+1} \Rightarrow \boxed{A=-10, B=20}$$



A uniform lamina is in the form of an isosceles triangle ABC in which $AC = 2a$ and angle $ABC = 90^\circ$. The point D on AB is such that the ratio $DB:AB = 1:k$. The point E on CB is such that DE is parallel to AC . The triangle DBE is removed from the lamina (see diagram).

- (a) Find, in terms of k , the distance of the centre of mass of the remaining lamina $ADEC$ from the midpoint of AC . [4]



$$Q: (0, a/3)$$

$$P: (0, a - \frac{2a}{3k})$$

$$\text{Area of } \triangle ABC = \frac{2a^2}{2} = a^2$$

$$\text{Area of } \triangle AED = \frac{(\frac{2a}{k})(\frac{a}{k})}{2} = \frac{a^2}{k^2}$$

$$\bar{y} \left(a^2 - \frac{a^2}{k^2} \right) (P) = \left(a^2 \right) \left(\frac{a}{3} \right) (P) + \left(\frac{a^2}{k^2} \right) \left(a - \frac{2a}{3k} \right) (P)$$

$$\bar{y} \left(1 - \frac{1}{k^2} \right) = \frac{a}{3} - \frac{1}{k^2} \left(a - \frac{2a}{3k} \right)$$

$$\bar{y} (k^2 - 1) = \frac{ak^2}{3} - \left(a - \frac{2a}{3k} \right)$$

$$\bar{y} (3)(k+1)(k-1) = ak^2 - 3a + \frac{2a}{k}$$

$$\bar{y} (3)(k+1)(k-1)(k) = ak^3 - 3ak + 2a$$

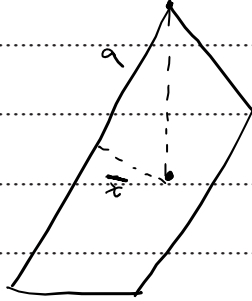
$$\bar{y} (3)(k+1)(k-1)(k) = a(k-1)(k^2+k-2)$$

$$\boxed{\bar{y} = \frac{a(k^2+k-2)}{3k(k+1)}}$$

When the lamina $ADEC$ is freely suspended from the vertex A , the edge AC makes an angle θ with the downward vertical, where $\tan \theta = \frac{5}{18}$.

(b) Find the value of k .

[3]



$$\frac{\bar{x}}{a} = \frac{5}{18}$$

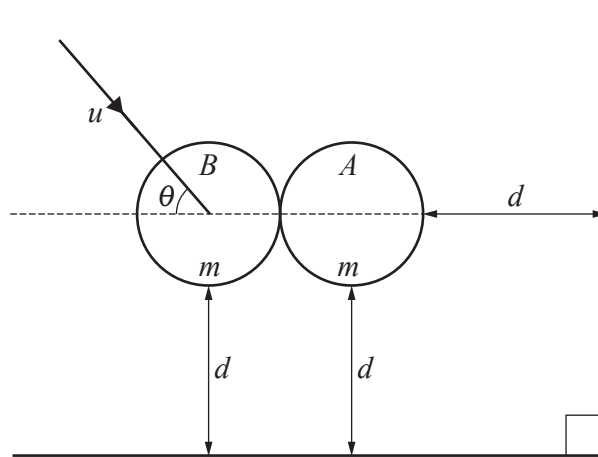
$$\frac{k^2+k-2}{3k(k+1)} = \frac{5}{18}$$

$$6k^2+6k-12 = 5k^2+5k$$

$$k^2+k-12=0$$

$$(k-3)(k+4)=0$$

$$\therefore \boxed{k=3}$$



Two smooth vertical walls meet at right angles. The smooth sphere A , with mass m , is at rest on a smooth horizontal surface and is at a distance d from each wall. An identical smooth sphere B is moving on the horizontal surface with speed u at an angle θ with the line of centres when the spheres collide (see diagram). After the collision, the spheres take the same time to reach a wall. The coefficient of restitution between the spheres is $\frac{1}{2}$.

(a) Find the value of $\tan \theta$.

[4]

→ Collisions only affect horizontal components.

$$\hookrightarrow mv \cos \theta = mv_{xB} + mv_{xA}$$

$$u \cos \theta = v_{xB} + v_{xA}$$

$$\hookrightarrow \frac{v_{xA} - v_{xB}}{u \cos \theta} = \frac{1}{2}$$

$$v_{xA} - v_{xB} = \frac{1}{2} u \cos \theta$$

$$\hookrightarrow v_{xA} = \frac{3}{4} u \cos \theta$$

$$v_{xB} = \frac{1}{4} u \cos \theta$$

Ball A only moves horizontally

$$\hookrightarrow t = \frac{d}{\frac{3}{4} u \cos \theta} = \frac{4d}{3u \cos \theta}$$

Ball B :

$$\begin{matrix} \frac{1}{4} u \cos \theta \\ \downarrow \\ u \sin \theta \end{matrix}$$

→ If Ball B collides with a wall at the same time as Ball A, it must be the lower wall as $v_{Ax} = 3v_{Bx}$, so A reaches much earlier

→ Time for B to reach lower wall :

$$t = \frac{d}{u \sin \theta}$$

$$\hookrightarrow \frac{d}{u \sin \theta} = \frac{4d}{3u \cos \theta}$$

$$\boxed{\tan \theta = \frac{3}{4}}$$

- (b) Find the percentage loss in the total kinetic energy of the spheres as a result of this collision. [3]

Initially:

$$KE = \frac{1}{2}mv^2$$

Finally:

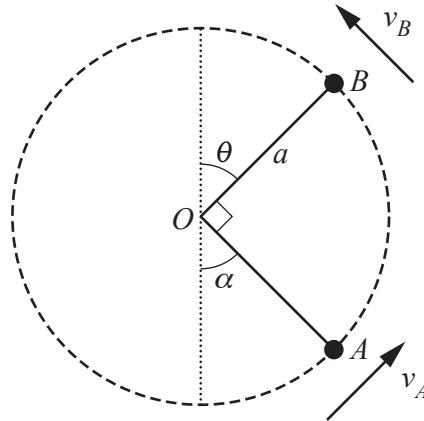
$$v_B = \sqrt{\left(\frac{1}{5}v\cos\theta\right)^2 + (v\sin\theta)^2} = \sqrt{\frac{v^2}{25} + \frac{9v^2}{25}} = \sqrt{\frac{10v^2}{25}}$$

$$v_A = \frac{3}{5}v\cos\theta = \frac{3}{5}v = \sqrt{\frac{9}{25}v^2}$$

$$\hookrightarrow KE = \frac{1}{2}m\left(\frac{9v^2}{25} + \frac{10v^2}{25}\right) = \frac{1}{2}m\left(\frac{19v^2}{25}\right)$$

$$= 0.38mv^2$$

$$\hookrightarrow \% \text{ loss} = \left(\frac{0.12mv^2}{0.5mv^2}\right) 100 = \boxed{24\%}$$



A bead of mass m moves on a smooth circular wire, with centre O and radius a , in a vertical plane. The bead has speed v_A when it is at the point A where OA makes an angle α with the downward vertical through O , and $\cos \alpha = \frac{3}{5}$. Subsequently the bead has speed v_B at the point B , where OB makes an angle θ with the upward vertical through O . Angle AOB is a right angle (see diagram). The reaction of the wire on the bead at B is in the direction OB and has magnitude equal to $\frac{1}{6}$ of the magnitude of the reaction when the bead is at A .

- (a) Find, in terms of m and g , the magnitude of the reaction at B . [6]

@A: $\Rightarrow$ where $v = \text{bottom speed}$

$$\cos \alpha = \frac{3}{5}$$

$$R = \frac{mv^2}{a} - 2mg + 3mg \cos \alpha$$

$$\sin \alpha = \frac{4}{5}$$

$$\tan \alpha = \frac{4}{3}$$

$$= \frac{mv^2}{a} - \frac{1}{5}mg$$

@B:

$$R = \frac{mv^2}{a} - 2mg + 3mg \cos(\alpha + 90)$$

$$= \frac{mv^2}{a} - 2mg - 3mg \sin \alpha$$

$$= \frac{mv^2}{a} - \frac{22}{5}mg$$

$$\hookrightarrow \left(\frac{mv^2}{a} - \frac{1}{5}mg \right) \left(\frac{1}{6} \right) = \frac{mv^2}{a} - \frac{22}{5}mg$$

$$\frac{v^2}{a} - \frac{1}{5}g = \frac{132}{5}g - \frac{6v^2}{a}$$

$$\frac{7v^2}{a} = \frac{133}{5}g$$

$$\hookrightarrow \frac{v^2}{a} = \frac{19}{5}g$$

$$T_B = \frac{mv^2}{a} - 2mg - 3mg \sin \alpha$$

$$= \frac{19}{5}mg - 2mg - \frac{12}{5}mg$$

$$= \frac{3}{5}mg \text{ towards center}$$

$$= \frac{3}{5}mg$$

- (b) Given that $v_A = \sqrt{kag}$, find the value of k . [2]

$$\frac{mv_A^2}{r} = \frac{mv^2}{r} - 2mg + 2mg \cos \alpha$$

$$\frac{mv_A^2}{r} = m \left(\frac{19}{50} \right) - \frac{4}{5} mg$$

$$v_A^2 = 3ag$$

$$v_A = \sqrt{3ag}$$

- 6 A particle P is projected with speed u at an angle α above the horizontal from a point O on a horizontal plane and moves freely under gravity. The horizontal and vertical displacements of P from O at a subsequent time t are denoted by x and y respectively.

(a) Derive the equation of the trajectory of P in the form

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha. \quad [3]$$

At a time 't':

Horizontally:

$$x = ut \cos \alpha$$

$$\hookrightarrow t = \frac{x}{u \cos \alpha}$$

Vertically:

$$y = (u \sin \alpha)(t) - \frac{1}{2}(g)(t^2)$$

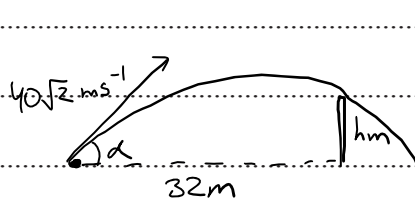
$$\hookrightarrow y = \frac{u \sin \alpha (x)}{u \cos \alpha} - \frac{1}{2}(g)\left(\frac{x^2}{u^2 \cos^2 \alpha}\right)$$

$$y = x \tan \alpha - \frac{gx^2}{2u^2} \sec^2 \alpha$$

During its flight, P must clear an obstacle of height h m that is at a horizontal distance of 32 m from the point of projection. When $u = 40\sqrt{2} \text{ m s}^{-1}$, P just clears the obstacle. When $u = 40 \text{ m s}^{-1}$, P only achieves 80% of the height required to clear the obstacle.

(b) Find the two possible values of h .

[6]

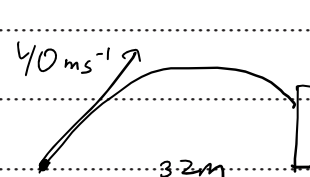


$$\text{subs } (x, y, u) = (32, h, 40\sqrt{2})$$

$$h = 32 \tan \alpha - 5 \left(\frac{32}{40\sqrt{2}} \right)^2 \sec^2 \alpha$$

$$h = 32 \tan \alpha - 1.6 \sec^2 \alpha$$

①



$$\text{subs } (x, y, u) = (32, 0.8h, 40)$$

$$0.8h = 32 \tan \alpha - 5 \left(\frac{32}{40} \right)^2 \sec^2 \alpha$$

$$0.8h = 32 \tan \alpha - 3.2 \sec^2 \alpha$$

②

$$0.8(32 \tan d - 1.6 \sec^2 d) = 32 \tan d - 3.2 \sec^2 d$$

$$25.6 \tan d - 1.28 \sec^2 d = 32 \tan d - 3.2 \sec^2 d$$

$$40 \tan d - 2 \sec^2 d = 50 \tan d - 5 \sec^2 d$$

$$3(1 + \tan^2 d) = 10 \tan d$$

$$3 \tan^2 d - 10 \tan d + 3 = 0$$

$$(3 \tan d - 1)(\tan d - 3) = 0$$

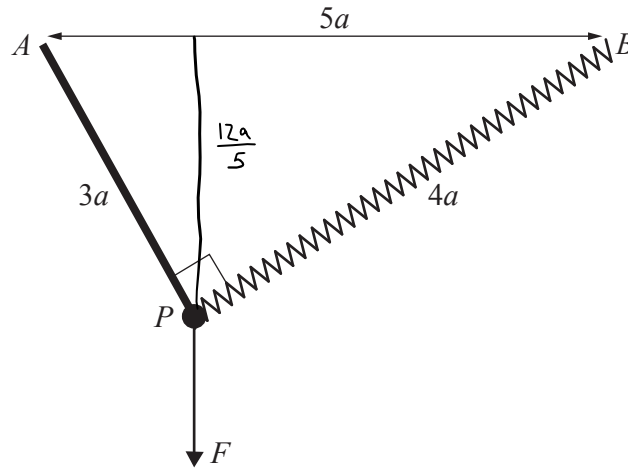
$$\tan d = \frac{1}{3} \quad \text{or} \quad 3$$

$$\sec^2 d = \frac{10}{9} \quad \text{or} \quad 10$$

$$h = 32 \tan d - 1.6 \sec^2 d$$

$$h = \frac{32}{3} - \left(\frac{8}{3}\right)\left(\frac{10}{9}\right) = \frac{80}{9}$$

$$\text{or } h = 32(3) - \left(\frac{3}{5}\right)(10) = 80$$



A particle P of mass m is attached to one end of a light rod of length $3a$. The other end of the rod is able to pivot smoothly about the fixed point A . The particle is also attached to one end of a light spring of natural length a and modulus of elasticity kmg . The other end of the spring is attached to a fixed point B . The points A and B are in a horizontal line, a distance $5a$ apart, and these two points and the rod are in a vertical plane.

Initially, P is held in equilibrium by a vertical force F with the stretched length of the spring equal to $4a$ (see diagram). The particle is released from rest in this position and has a speed of $\frac{6}{5}\sqrt{2ag}$ when the rod becomes horizontal.

(a) Find the value of k .

[5]

Initially :

$$EPE = \frac{\lambda x^2}{2l} = \frac{(kmg)(3a)^2}{2a} = \frac{9}{2} kmga$$

$$KE = 0$$

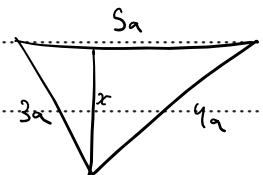
$$PE = 0$$

Finally :

$$EPE = \frac{\lambda x^2}{2l} = \frac{kmg(a)^2}{2a} = \frac{1}{2} kmga$$

$$KE = \frac{1}{2} m (v)^2 = \frac{1}{2} m \left(\frac{36}{25} \right) (2ag) = \frac{36}{25} mga$$

$$PE = mg \left(\frac{12a}{5} \right) = \frac{12}{5} mga$$



$$\frac{3a}{x} = \frac{5a}{4a}$$

$$\Rightarrow x = \frac{12a}{5}$$

By CoE:

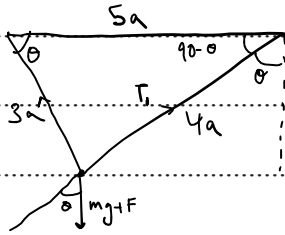
$$\frac{9}{2} kmga = \frac{1}{2} kmga + \frac{12}{5} mga + \frac{36}{25} mga$$

$$4k = 3.84$$

$$k = 0.96$$

(b) Find F in terms of m and g .

[2]



$$T_1 = (mg + F) \cos \theta$$

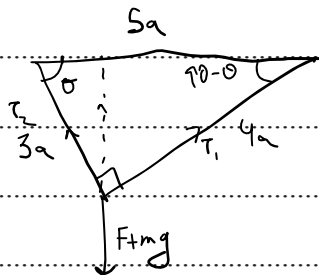
$$\frac{T_2}{1} = (mg + F) \left(\frac{3}{5} \right)$$

$$\left(\frac{0.96mg(3a)}{a} \right) \left(\frac{5}{3} \right) - mg = F$$

$$F = 3.8mg$$

(c) Find, in terms of m and g , the tension in the rod immediately before it is released.

[2]



$$F + mg = T_1 \sin(90 - \theta) + T_2 \sin \theta$$

$$4.8mg = \frac{(0.96mg)(3a)}{a} \left(\frac{3}{5} \right) + T_2 \left(\frac{4}{5} \right)$$

$$24mg = 8.64mg + 4T_2$$

$$T_2 = 3.84mg$$

[illegible]

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